

Astra and Opus

Five challenging parts

Prepared by Ryan | 7 October 2026

Independent block layouts and realized hexahedral meshes for five selected CAD geometries. Each model received the same geometry, layout preferences and fixed acceptance checks.

GPT 6 Astra

5 / 5 counted

\$15.07 API equivalent | 61.55 min

Opus 5.5

4 / 5 counted

\$31.97 CLI usage | 176.72 min

All ten final meshes pass geometry checks.

One Opus attempt remains rejected by the fixed command audit. Its logs indicate an apparent false positive from echoed command text; the rejection has been retained. Each model used 10 full mesh revisions in total.

HOW TO READ THIS REPORT

Updated 9 October 2026 with optimised layouts. Page 2 compares every layout's worst element before and after the corner optimiser (hexlayoutopt), which keeps the model's blocks and moves only their corners. Each part then has one page per model: the drawn layout with its top-down and isometric meshes, and below it the optimised layout with its own meshes. The optimised layout replaces the drawn one only if every check passes and its worst element is no lower: 8 of 10 improved and 2 are kept as drawn. SJ (scaled Jacobian) measures element quality; larger is better, with a fixed minimum of 0.2; colors share the range 0.2 to 1 (low yellow, high lavender) and show exposed surfaces.

Selected showcase, not a population benchmark: these geometries had previously verified meshes, which were withheld from authors. Astra used medium reasoning effort; Opus used high. Usage figures have different accounting bases and are not subscription invoices. No human quality grades were assigned.

Worst element before and after the corner optimiser

Each line is one layout: the model's drawn layout (hollow gray) and the same blocks with their corners moved by the optimiser (filled). Teal: accepted, replacing the drawn layout. Amber: the optimised layout failed a 3-D check, so the drawn layout stands. Min SJ is the worst element's scaled Jacobian in the nominal mesh; higher is better, 0.2 is the floor.



8 of 10 layouts improved, 0 lost. Median change across all ten, counting the two kept as drawn as 0: +0.101; largest +0.206. Every accepted layout passes the same frozen checks as the drawn one and stays at least 0.20 with the element counts doubled. The two rejected layouts raised the worst element but pushed one CAD face's elements past the 3-D aspect-ratio limit of 20.

Rounded plate with bores and slot

t8_profile_01098d50

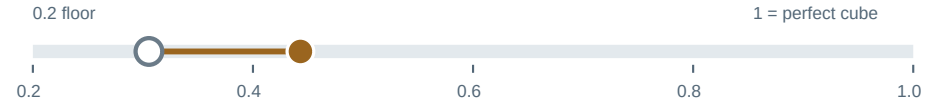
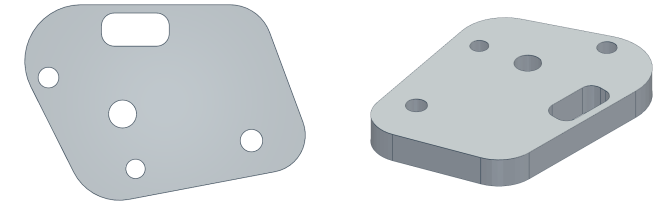
WORST ELEMENT (MIN SJ)

0.3055 → **0.4431**

not accepted: drawn layout kept

The optimised layout raised the worst element but one CAD face's elements reached aspect ratio 27.1 against the 3-D limit of 20 (drawn: 19.6), so it is not used.

ORIGINAL CAD



DRAWN

by GPT 6 Astra

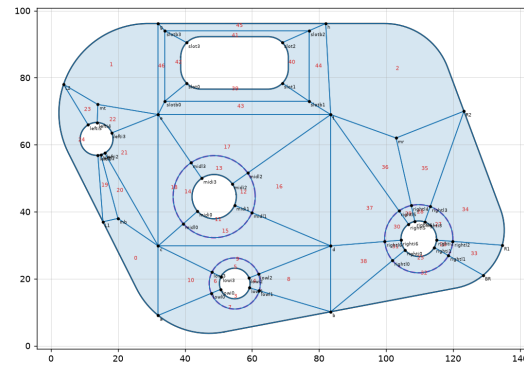
47 blocks | 5,086 hexes

Min SJ: 0.3055

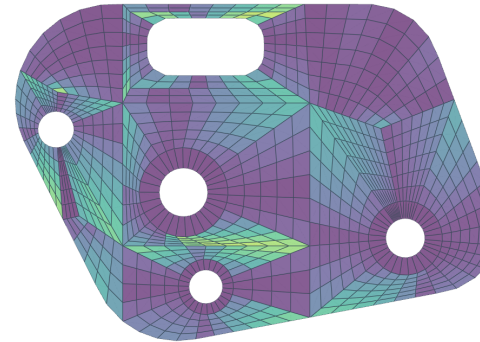
Doubled SJ: 0.3055

Counted pass

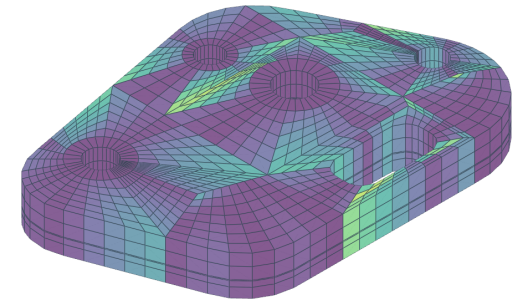
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

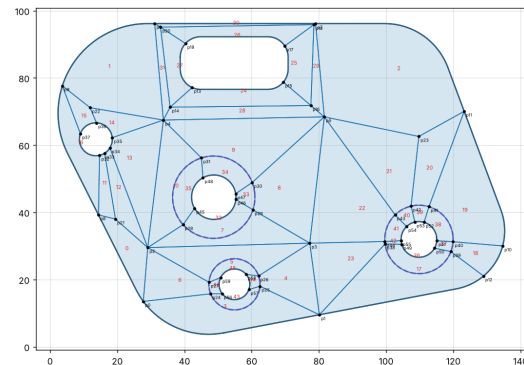
47 blocks | 5,214 hexes

Min SJ: 0.4431

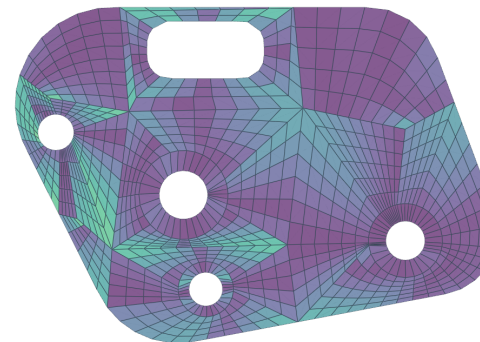
Doubled SJ: 0.4648

Not accepted (3-D aspect check)

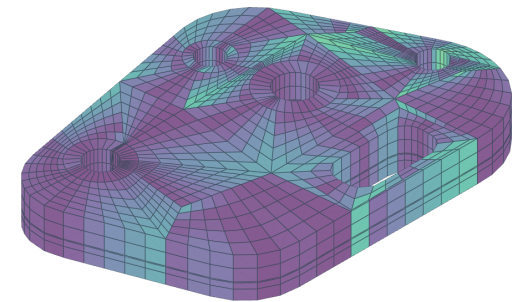
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



0.2 floor

SJ color scale (all meshes)



Exposed faces colored by cell SJ; interior cells included in metrics.

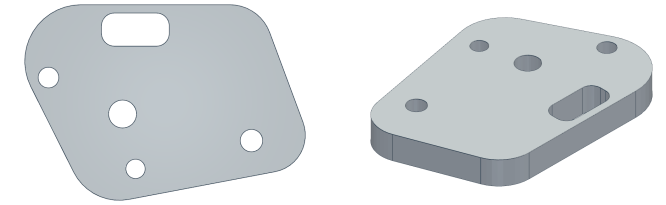
Rounded plate with bores and slot

t8_profile_01098d50

WORST ELEMENT (MIN SJ)

0.5661 → **0.6864**
+0.120

ORIGINAL CAD



DRAWN

by Opus 5.5

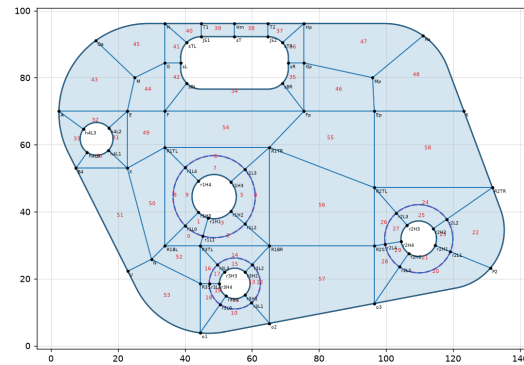
59 blocks | 4,484 hexes

Min SJ: 0.5661

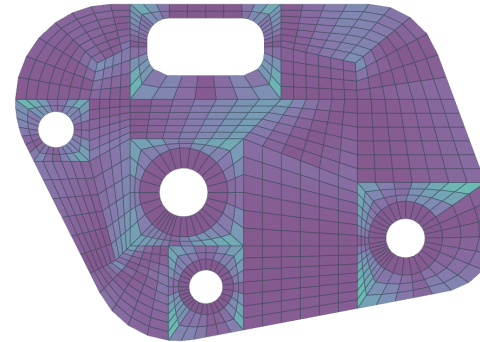
Doubled SJ: 0.5661

Counted pass

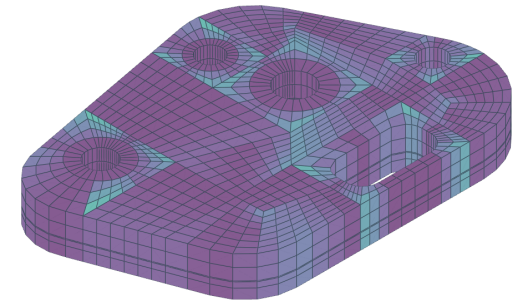
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

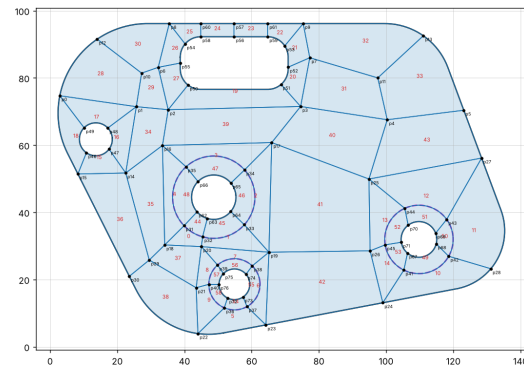
59 blocks | 4,966 hexes

Min SJ: 0.6864

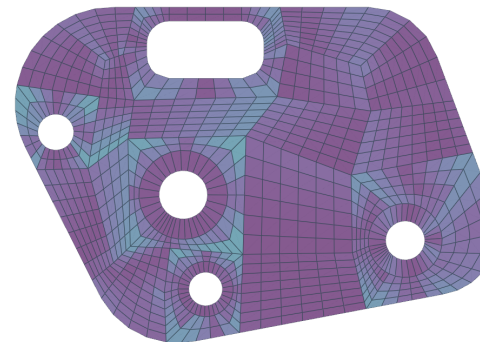
Doubled SJ: 0.6864

Accepted: replaces the drawn layout

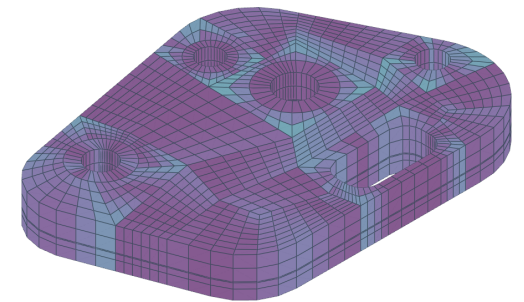
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



SJ color scale (all meshes)



Exposed faces colored by cell SJ; interior cells included in metrics.

Notched plate with stepped features

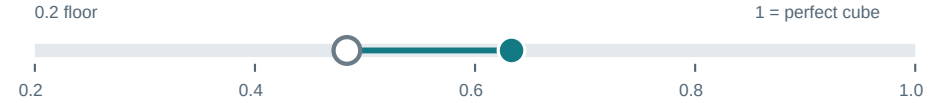
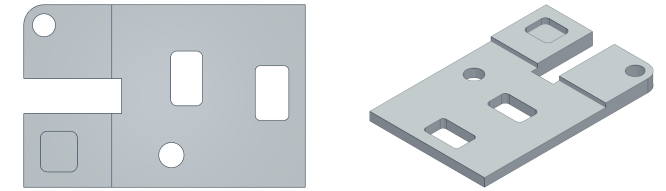
t8_profile_011291e8

WORST ELEMENT (MIN SJ)

0.4835 → 0.6332

+0.150

ORIGINAL CAD



DRAWN

by GPT 6 Astra

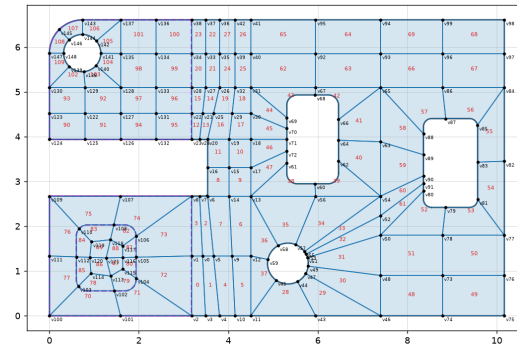
110 blocks | 2,332 hexes

Min SJ: 0.4835

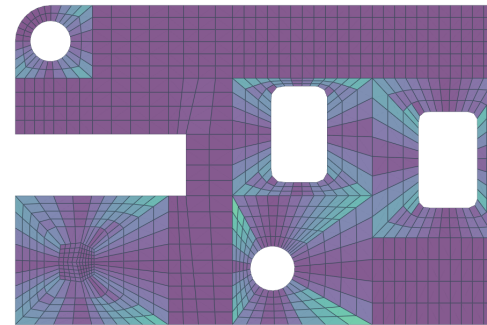
Doubled SJ: 0.4835

Counted pass

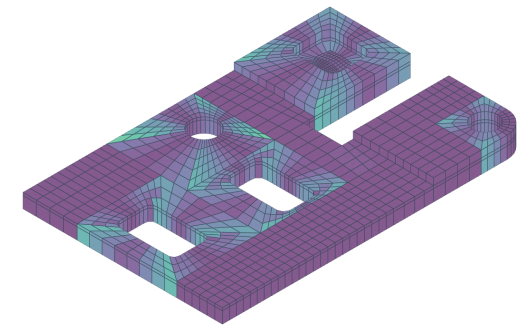
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

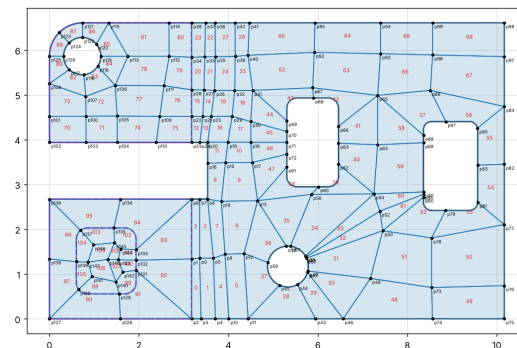
110 blocks | 2,582 hexes

Min SJ: 0.6332

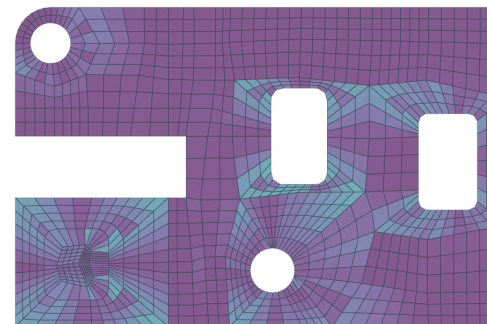
Doubled SJ: 0.6332

Accepted: replaces the drawn layout

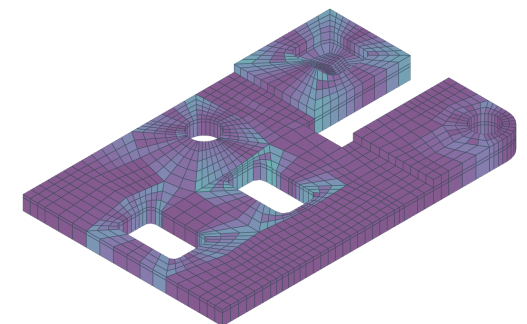
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



SJ color scale (all meshes)

0.2 floor



Exposed faces colored by cell SJ; interior cells included in metrics.

Notched plate with stepped features

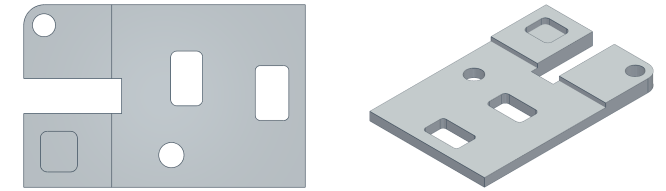
t8_profile_011291e8

WORST ELEMENT (MIN SJ)

0.4754 → **0.6815**

+0.206

ORIGINAL CAD



DRAWN

by Opus 5.5

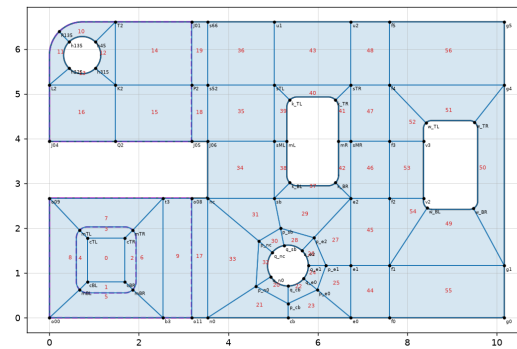
57 blocks | 1,698 hexes

Min SJ: 0.4754

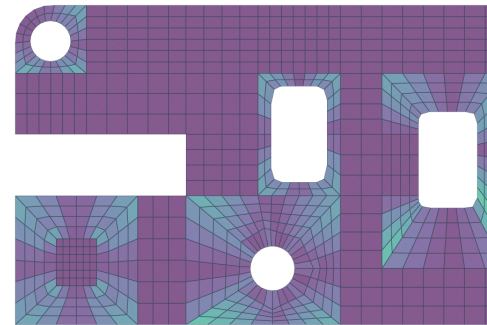
Doubled SJ: 0.4754

Counted pass

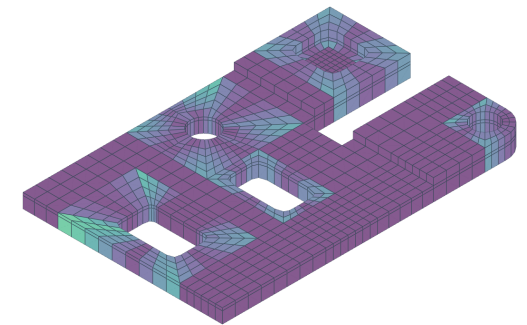
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

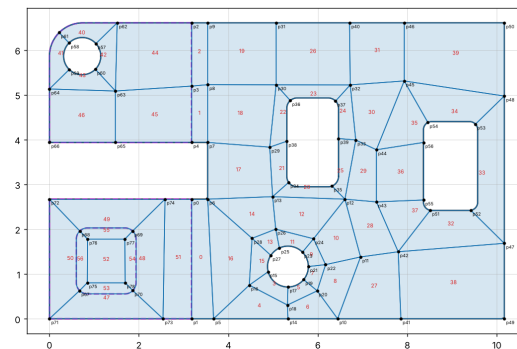
57 blocks | 1,895 hexes

Min SJ: 0.6815

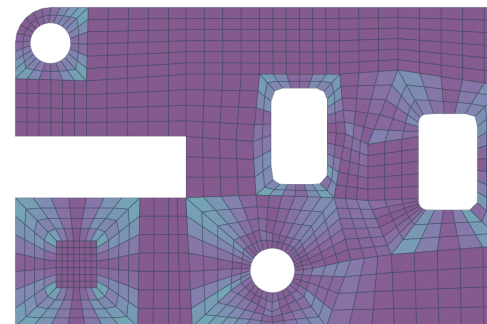
Doubled SJ: 0.6815

Accepted: replaces the drawn layout

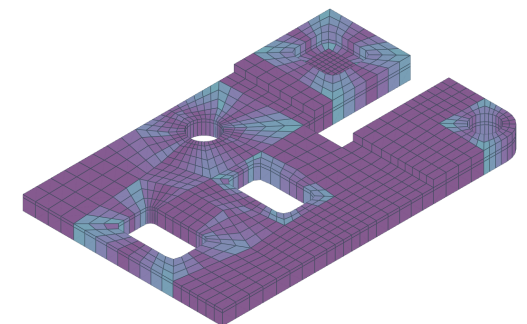
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



SJ color scale (all meshes)



Exposed faces colored by cell SJ; interior cells included in metrics.

Stepped hexagonal body

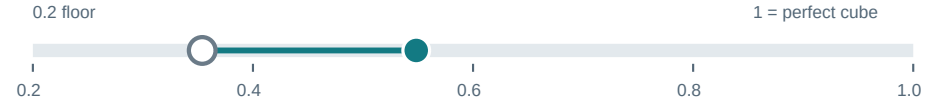
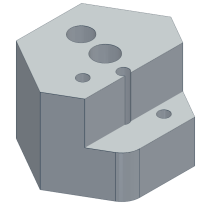
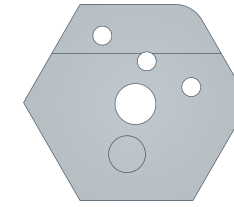
t8_block_0403c4d8

WORST ELEMENT (MIN SJ)

0.3539 → **0.5484**

+0.194

ORIGINAL CAD



DRAWN

by GPT 6 Astra

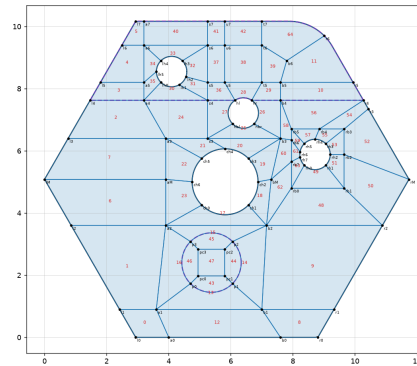
65 blocks | 11,068 hexes

Min SJ: 0.3539

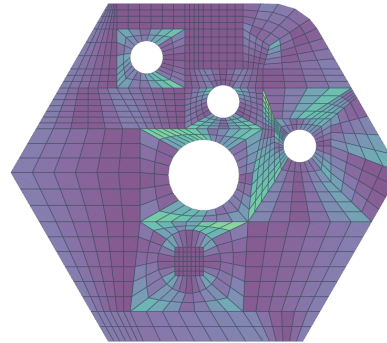
Doubled SJ: 0.3539

Counted pass

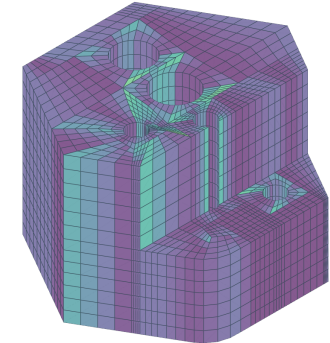
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

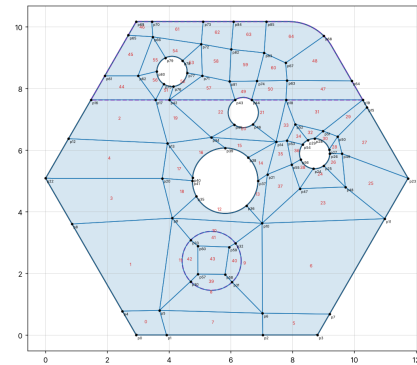
65 blocks | 12,240 hexes

Min SJ: 0.5484

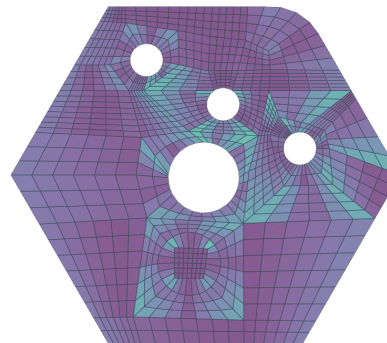
Doubled SJ: 0.5484

Accepted: replaces the drawn layout

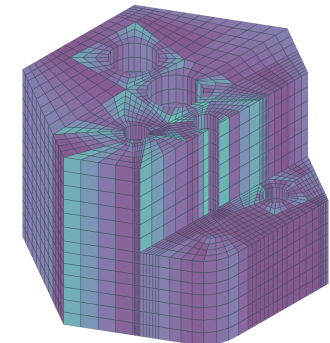
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



SJ color scale (all meshes)

0.2 floor

0.2

0.5

1

Exposed faces colored by cell SJ; interior cells included in metrics.

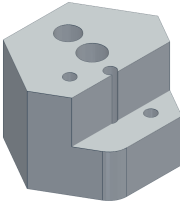
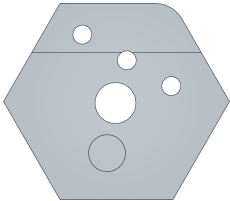
Stepped hexagonal body

t8_block_0403c4d8

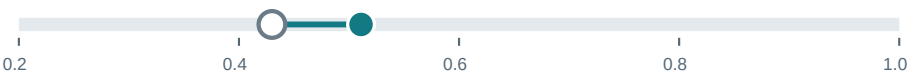
WORST ELEMENT (MIN SJ)

0.4300 → 0.5109 +0.081

ORIGINAL CAD



0.2 floor



DRAWN

by Opus 5.5

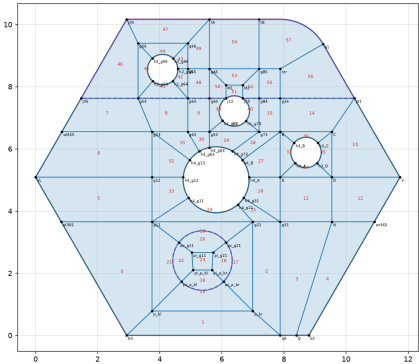
58 blocks | 13,724 hexes

Min SJ: 0.4300

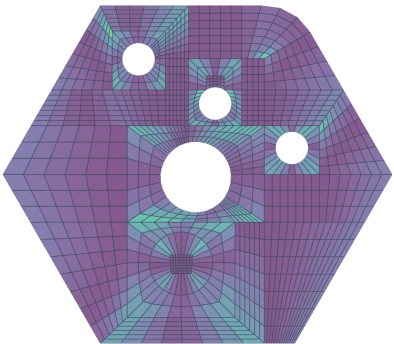
Doubled SJ: 0.4300

Geometry pass / audit rejected

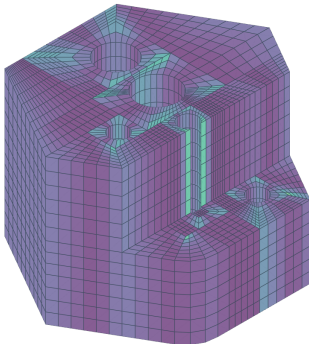
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

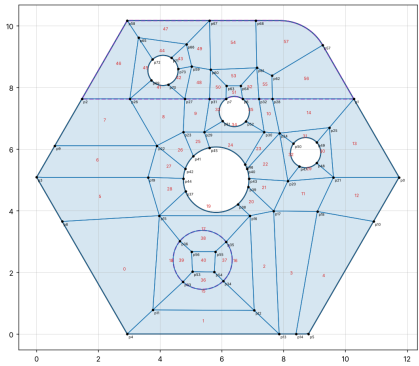
58 blocks | 13,890 hexes

Min SJ: 0.5109

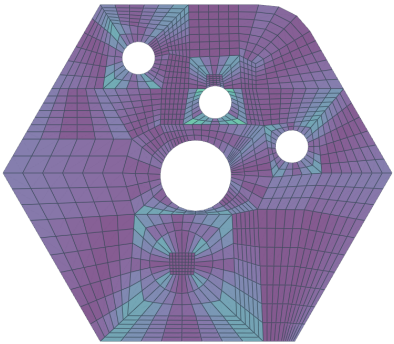
Doubled SJ: 0.5113

Accepted: replaces the drawn layout

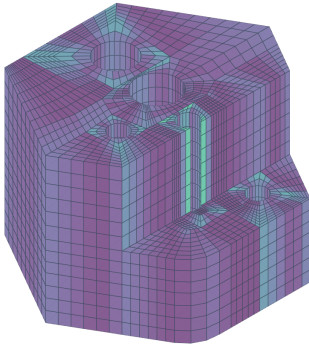
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



0.2 floor

SJ color scale (all meshes)



Exposed faces colored by cell SJ; interior cells included in metrics.

Rounded triangular plate

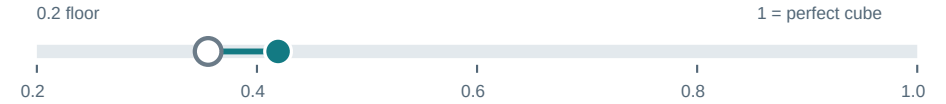
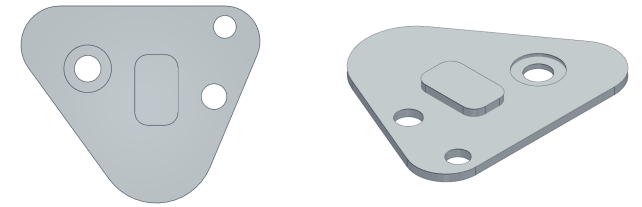
t8_profile_01040740

WORST ELEMENT (MIN SJ)

0.3557 → **0.4192**

+0.064

ORIGINAL CAD



DRAWN

by GPT 6 Astra

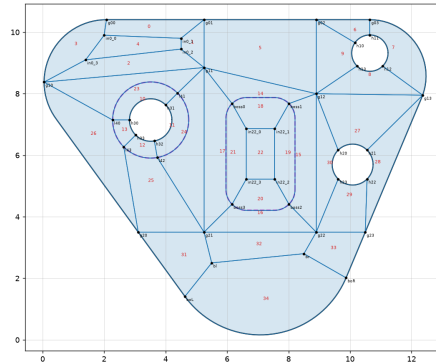
35 blocks | 2,244 hexes

Min SJ: 0.3557

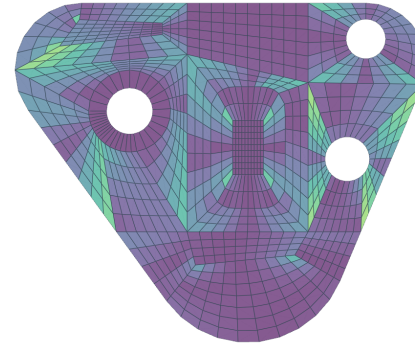
Doubled SJ: 0.3557

Counted pass

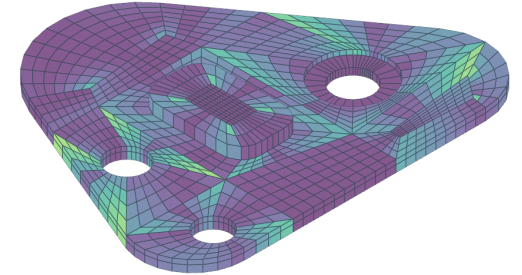
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

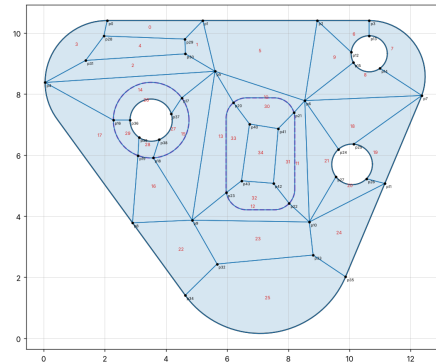
35 blocks | 2,438 hexes

Min SJ: 0.4192

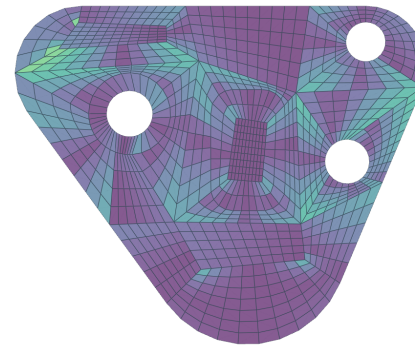
Doubled SJ: 0.4192

Accepted: replaces the drawn layout

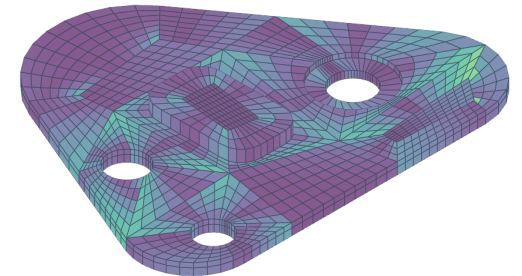
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



SJ color scale (all meshes)



Exposed faces colored by cell SJ; interior cells included in metrics.

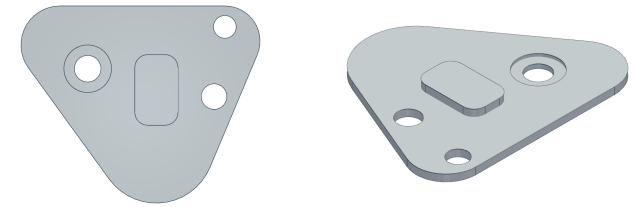
Rounded triangular plate

t8_profile_01040740

WORST ELEMENT (MIN SJ)

0.4167 → **0.5809**
+0.164

ORIGINAL CAD



DRAWN

by Opus 5.5

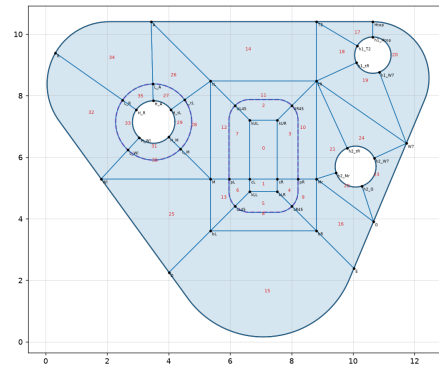
36 blocks | 2,152 hexes

Min SJ: 0.4167

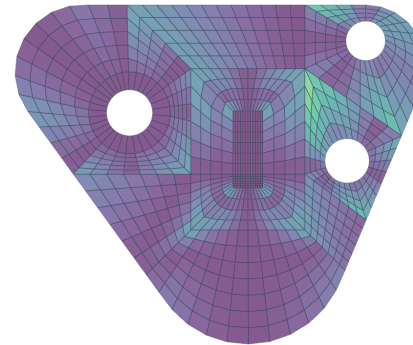
Doubled SJ: 0.4167

Counted pass

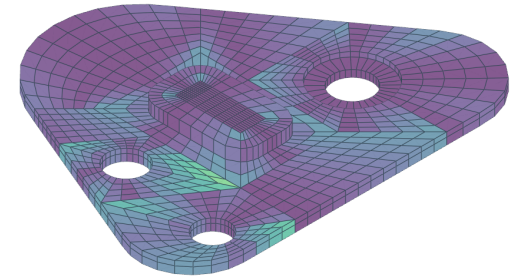
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

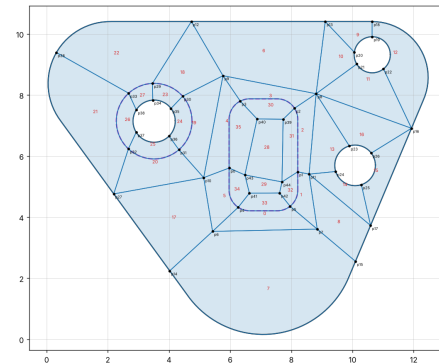
36 blocks | 2,227 hexes

Min SJ: 0.5809

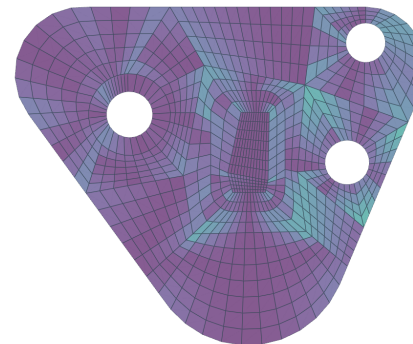
Doubled SJ: 0.5809

Accepted: replaces the drawn layout

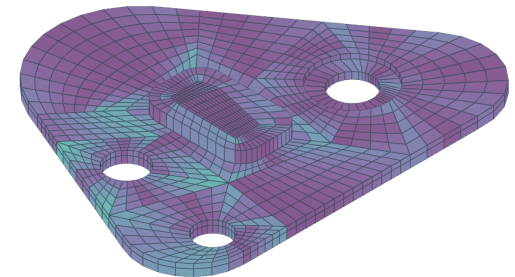
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



SJ color scale (all meshes)



Exposed faces colored by cell SJ; interior cells included in metrics.

Stepped block with pockets

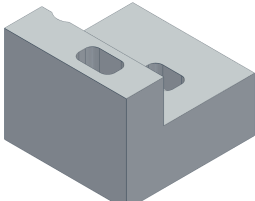
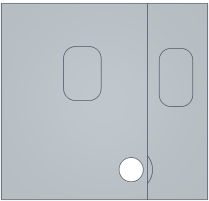
t8_block_040566d0

WORST ELEMENT (MIN SJ)

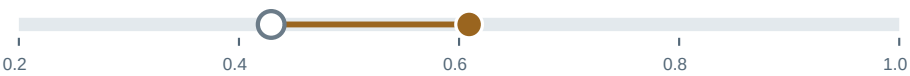
0.4293 → 0.6092 not accepted: drawn layout kept

The optimised layout raised the worst element but one CAD face's elements reached aspect ratio 21.5 against the 3-D limit of 20 (drawn: 17.4), so it is not used.

ORIGINAL CAD



0.2 floor



DRAWN

by GPT 6 Astra

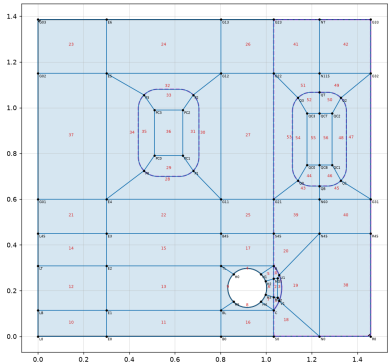
57 blocks | 7,874 hexes

Min SJ: 0.4293

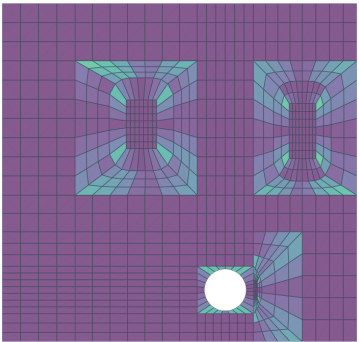
Doubled SJ: 0.4293

Counted pass

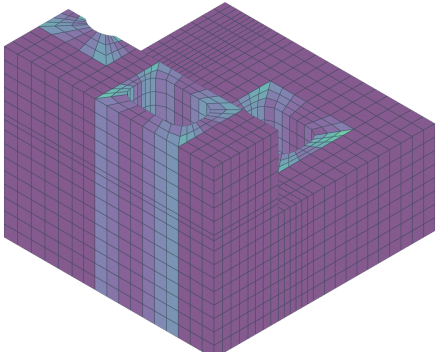
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

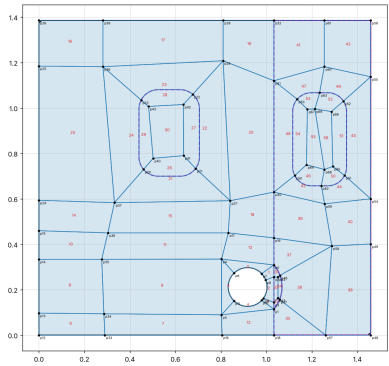
57 blocks | 8,713 hexes

Min SJ: 0.6092

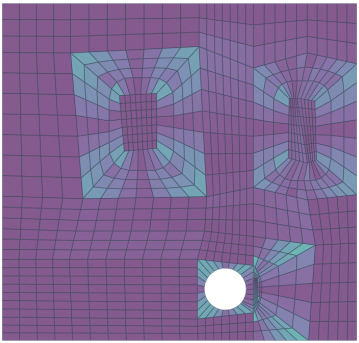
Doubled SJ: 0.6092

Not accepted (3-D aspect check)

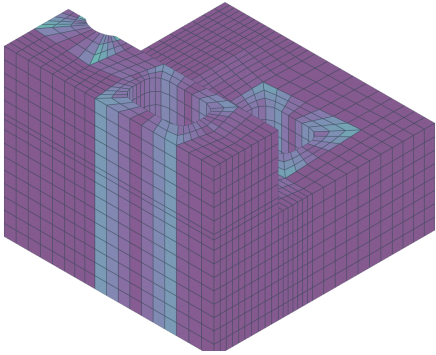
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



0.2 floor

SJ color scale (all meshes)



Exposed faces colored by cell SJ; interior cells included in metrics.

Stepped block with pockets

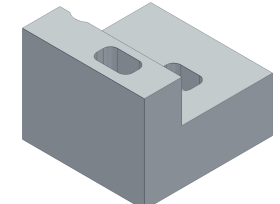
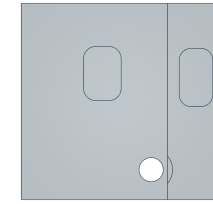
t8_block_040566d0

WORST ELEMENT (MIN SJ)

0.5464 → 0.5754

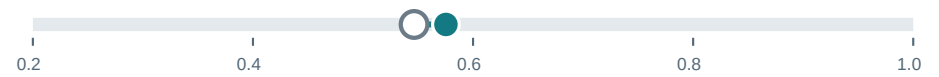
+0.029

ORIGINAL CAD



0.2 floor

1 = perfect cube



DRAWN

by Opus 5.5

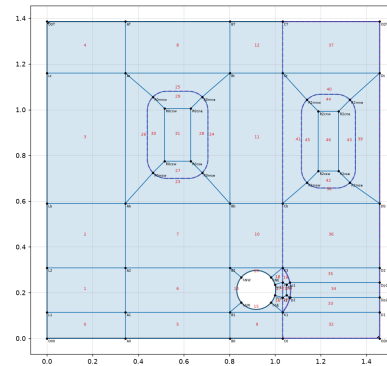
47 blocks | 6,902 hexes

Min SJ: 0.5464

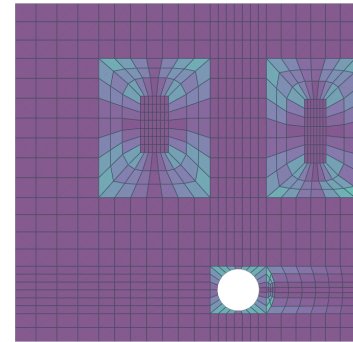
Doubled SJ: 0.5745

Counted pass

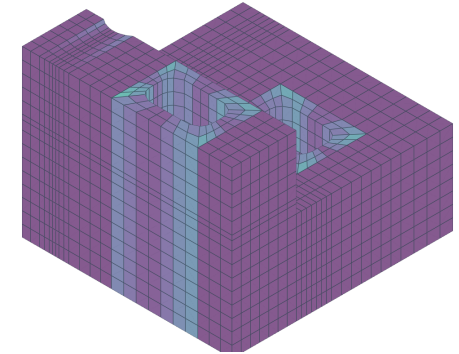
BLOCK LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



OPTIMISED

same blocks, corners moved

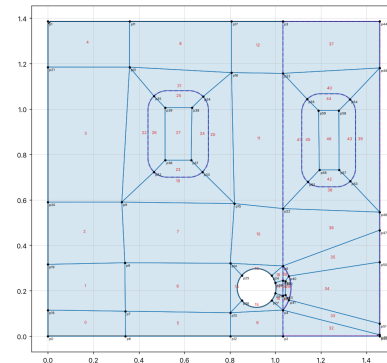
47 blocks | 6,901 hexes

Min SJ: 0.5754

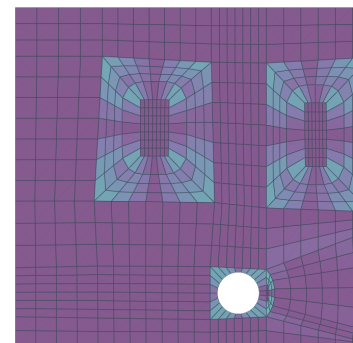
Doubled SJ: 0.5758

Accepted: replaces the drawn layout

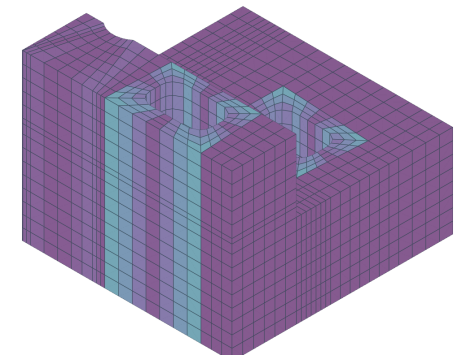
OPTIMISED LAYOUT



TOP-DOWN MESH



ISOMETRIC MESH



0.2 floor

SJ color scale (all meshes)



Exposed faces colored by cell SJ; interior cells included in metrics.